Akash Goswami

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Education

NIT Warangal **May 2027**
Bachelor of Technology in Computer Science Telangana, India

Experience

Software Developer Intern **June – August 2026**
AVEVA Hyderabad, India

- Designed and built **DocAnalytics**, a multi-tenant document-processing analytics platform, end-to-end from data modeling to deployment. Hosted in production under a custom domain.
- Built the core platform and **RESTful APIs** using **.NET 10**, **Angular 22** with Signals, and **PostgreSQL** via EF Core, enforcing strict multi-tenant data isolation through global query filters and role-based authorization.
- Shipped an **AI-powered invoice-extraction pipeline** using **AWS Bedrock** and **S3**, achieving ∼**98%** extraction confidence via automated subtotal/total reconciliation, and hardened it with AWS GuardDuty malware scanning.
- Provisioned and deployed production **AWS** infrastructure using **Terraform** (**ECS**, **ALB**, **RDS**, **ECR**), behind a **CI/CD** pipeline gated on test coverage, gitleaks, and Trivy scans, with API rate limiting.
- Applied **Test-Driven Development (TDD)** using **xUnit** and **Vitest**, and load-tested the API with **JMeter**, confirming sub-3-second response times with **0% errors** at 1M+ record scale.

Projects

Twizz | *Python, Tesseract OCR, Docker, AWS EC2* **September 2025**

- Developed an **automated Discord bot** that identifies and delivers job postings to over **7000 users across 4 continents** by scraping sources such as LinkedIn, Reddit threads, and other online platforms.
- Containerized the alert delivery system using **Docker**, enabling portable deployments while optimizing performance with session caching, asynchronous processing, and automated resource cleanup.
- Hosted the bot on **AWS EC2** in a headless setup, leveraging **AWS CloudWatch** and **Lambda** for monitoring and automated recovery, improving data durability and reducing downtime.

Investment Portfolio Management System | *SQL, Database Architecture* **July 2025**

- Structured over **12 normalized, BCNF-compliant** SQL tables with strong referential integrity to efficiently track transactions, holdings, and returns across asset classes.
- Developed advanced **SQL** queries for FIFO stock sales, SIP NAV aggregation, and return projections, enabling real-time portfolio insights and performance tracking.

CryptoHunt - Linux Emulator | *Node.js, MongoDB, Heroku* **November 2024**

- Developed a **Linux emulator** for an online Capture-The-Flag competition at Technozion '24, featuring multi-level challenges and hidden flags.
- Built a virtual filesystem with core Linux commands (**ls,cat,grep**) and optimized performance by converting level structures into **JSON**, reducing database queries by **40%** and improving response time by **25%**.
- Implemented backend logic in Node.js for command parsing, filesystem state management, and deployed the platform on Heroku for seamless online access.

Technical Skills

Languages: C++, C#, C, Python, Java, TypeScript, SQL
Frameworks/Libraries: .NET 10, ASP.NET Core, Angular 22, EF Core
Databases: PostgreSQL, Oracle SQL
Cloud & DevOps: AWS (S3, RDS, Bedrock, ECS, IAM, GuardDuty), Terraform, Docker, CI/CD
Tools: Git, GitHub, Swagger, Postman, JMeter, xUnit, Vitest, Visual Studio, Google Colab

Leadership

Additional Secretary, CSE Society, NITW **Sept 2024 – Present**

- Led the Machine Learning Team and co-led **Noobathon 5.0**, a 24-hour hackathon for first-years.
- Built **Crypto-Hunt-Online**, a Linux-emulator CTF project showcased at Technozion '24.

Achievements

- Earned the **Super Contributor Badge** for **6+ Open-Source** PRs merged in **Hacktoberfest 2025**.
- Secured **2nd place** at **HackQE Hackathon 2021** for OxyMap, a COVID-19 resource locator.
- Awarded **$100** grant by **HackClub** for designing a custom PCB board, with the design open-sourced on GitHub.